

Modeling and Control of Complex-shaped VTOL Systems Including Aerodynamic Forces

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Abstract—This paper presents a modeling and control framework for complex-shaped VTOL vehicles flying in presence of non-negligible aerodynamic effects. First, aerodynamic forces are calculated during robot flight in different operating conditions by means of Computational Fluid Dynamics (CFD) analysis. Then, analytical models of the aerodynamics coefficients are generated from the dataset collected with CFD analysis, and used to improve the flying robot control design. We present two control strategies: compensate for the aerodynamic effects via feedback linearization and enforcing the controller robustness with gain scheduling. Simulation results on the jet-powered humanoid robot iRonCub validate the proposed approach.

I. INTRODUCTION

Recent researches on Vertical Take Off and Landing (VTOL) systems have shown interest on both *passive* (e.g., inspection and remote sensing) and *active* tasks (e.g., grasping and manipulation). The dexterity required for achieving the latter often calls for the design of complex-shaped aerial vehicles with several Degrees of Freedom (DoF), which complicate the control design [1] [2] [3].

Additionally, when the VTOL needs to operate in outdoor environments, the control design is further complicated by the environment uncertainties, e.g. the aerodynamic effects that arise in presence of wind and while the vehicle is performing high speed movements [4] [5]. In this paper, we propose a simple yet effective modeling and control strategy to handle these aerodynamic effects on a complex-shaped, high DoF VTOL system: a jet-powered humanoid robot.

The combination of terrestrial and aerial locomotion in a single robotic platform is of some interest in the robotics community, which recently proposed hexapod-quadrotor designs, hybrid terrestrials and aerial quadrotors, insect biobots, and humanoid robots with thrusters [6] [7] [8] [9] [10]. An attempt to unify manipulation, aerial, and bipedal terrestrial locomotion on a single robotic platform is being carried out by the iRonCub project [11] [12] [13]. As shown in Fig. 1, four jet engines located on the robot arms and chest are used to lift the robot from the ground. The increased operativity and mobility of this type of platforms, however, often come at the cost of a complex-shaped hardware design with numerous DoF.

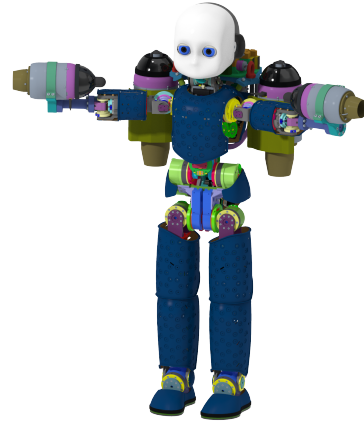


Fig. 1. CAD model of the jet-powered humanoid robot iRonCub.

A common assumption for high DoF VTOL systems is that the vehicle operates at relatively *small speed*, often in indoor environments [1] [2] [3], correspondingly the aerodynamic forces acting on the system are usually negligible and not considered in modeling and control framework [4] [19]. When aerodynamic effects are important enough to affect system stability during flight, or high precision is required for performing a task, different modeling and control strategies could be adopted, e.g. to estimate aerodynamic force by designing a momentum-based observer or other disturbances estimation strategies [24] [23], and to compensate estimated aerodynamic force by means of adaptive control strategies [21] [22]. However, the absence of an explicit model of the aerodynamic forces could impair the effectiveness of these methods for exploiting physical properties that are beneficial for flight.

High-speed flight can benefit from the presence of lift forces that alleviate the effort made by the propellers for gravity compensation. The VTOL dynamical model, in this case, explicitly includes aerodynamic forces [25]. Few examples include the analysis of blade flapping on quadrotors, modeling first-order aerodynamic effects and evaluating the effect of drag force on thrust power under hovering, [18] [27] [26]. However, the shape of aerial vehicles has a significant role in modeling aerodynamics effects. It is difficult to design an effective aerodynamic model in case of complex-shaped, high DoF aerial systems [17] [25].

Different techniques have been developed to properly estimate aerodynamic forces for complex-shaped systems, e.g. wind tunnel test and Computational Fluid Dynamics

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(CFD) analysis. However, to estimate the aerodynamic forces for an entire set of robot operating conditions, i.e. *flight envelope*, requires to collect a very large number of data. An alternative is to design simpler analytical models via limited dataset provided by wind tunnel tests or CFD simulations. Many of these analytical models assume that the robot body is axisymmetric [14] [30] [32].

In this paper we propose a framework to include aerodynamic forces in the modeling and control design of high DoF VTOL systems, which in the case under study are represented by the flying humanoid robot iRonCub. To improve the iRonCub operativity in presence of aerodynamic effects, we applied the following procedure. At first, a series of CFD simulations is performed on a simplified robot model, in order to generate a dataset of the aerodynamic forces acting on the robot for a given flight envelope. Secondly, a model for a non-axisymmetric bluff body is designed to fit the collected dataset and estimate the aerodynamic characteristics in between the flight envelope. The model is verified to be also physically consistent (positivity of the Drag coefficient). Finally, the designed model is used to improve the iRonCub flight-control strategy developed in our previous works [12] [13] by compensation of the aerodynamic forces via *feedback linearization* and improvement of the controller robustness with *gain scheduling*.

The paper is organized as follows. Sec. II recalls notation, system modeling and fundamentals of aerodynamic modeling. In Sec. III, we describe the CFD analysis on a simplified iRonCub model and the identification of the aerodynamic characteristics. Sec. IV presents the improvements of iRonCub flight controller to handle aerodynamic effects. Simulation results are displayed in Sec. V. Sec. VI concludes the paper and points out future directions.

II. BACKGROUND

A. Notation

- $\mathcal{I} = \{O; \mathbf{X}, \mathbf{Y}, \mathbf{Z}\}$ denotes an inertial frame, composed of a point (origin) and an orientation as the ENU (East-North-Up) convention;
- $\mathcal{B}$ denotes the *base frame*, i.e. a frame rigidly attached to the robot *base link*;
- $\mathcal{G}[\mathcal{I}]$ is a frame with the origin at the robot center of mass and the same orientation of the inertial frame;
- ${}^{\mathcal{I}}o_G \in \mathbb{R}^3$ is the Center of Mass (CoM) position of the robot in the inertial frame. ${}^{\mathcal{I}}v_{CoM} \in \mathbb{R}^3$ denotes the linear velocity of the robot CoM w.r.t. the inertial frame;
- ${}^{\mathcal{I}}v_a \in \mathbb{R}^3$ indicates the relative velocity between the wind and the robot CoM expressed in the inertial frame, i.e. ${}^{\mathcal{I}}v_a = {}^{\mathcal{I}}v_{CoM} - {}^{\mathcal{I}}v_w$, with ${}^{\mathcal{I}}v_w \in \mathbb{R}^3$ the absolute wind velocity¹;
- $\mathcal{D} = \{{}^{\mathcal{I}}o_G; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ represents the *body frame*. The $\mathbf{k}$ axis direction is parallel to the chest cylinder principal axis, pointing towards the legs. The $\mathbf{j}$ axis is normal

¹In the remainder of the paper, we omit the superscript $\mathcal{I}$ unless it is necessary to explicitly write it for the sake of clarity, and we write ${}^{\mathcal{I}}v_a = v_a$.

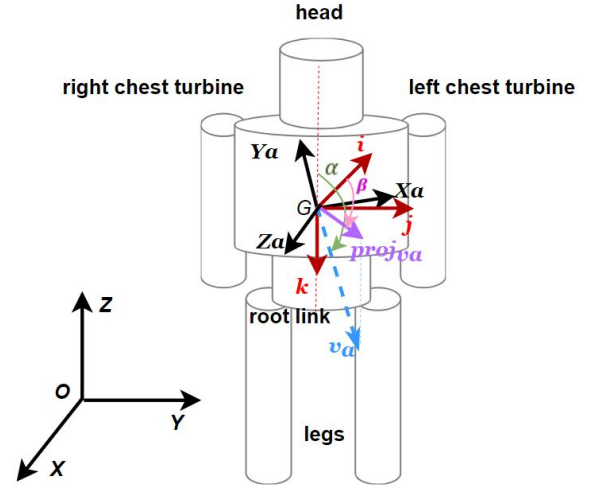


Fig. 2. Notation representation: inertial frame, body frame $\mathcal{D}$ (red color), relative velocity frame $\mathcal{A}$ (black color, on the robot), and (α, β) angles.

to the robot symmetry plane, pointing towards left arm. The $\mathbf{i}$ axis is defined by the right-hand rule;

- $\mathcal{A} = \{{}^{\mathcal{I}}o_G; \mathbf{X}_a, \mathbf{Y}_a, \mathbf{Z}_a\}$ is the *relative velocity frame*, with the $\mathbf{Y}_a$ axis along the opposite direction of the relative velocity ${}^{\mathcal{I}}v_a$.

B. Robot Modeling

The flying humanoid robot is modeled as a multi-body, *floating-base* system composed of $n + 1$ rigid bodies (*links*). Each body is connected to the others by means of 1-DoF joints, for a total of n DoF. The set of robot configurations belongs to a space defined as: $\mathbb{Q} \in \mathbb{R}^3 \times SO(3) \times \mathbb{R}^n$. The triplet $q = ({}^{\mathcal{I}}o_B, {}^{\mathcal{I}}R_B, s) \in \mathbb{Q}$ is composed of $({}^{\mathcal{I}}o_B, {}^{\mathcal{I}}R_B)$, which represent the position and orientation of the base frame $\mathcal{B}$ w.r.t. the inertial frame, and the robot internal shape $s \in \mathbb{R}^n$ represented by the joint angles. The system velocities belong to the space $\mathbb{V} \in \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^n$. An element of $\mathbb{V}$ is given by $v = (v_B, \dot{s})$ where $v_B = ({}^{\mathcal{I}}\dot{o}_B, {}^{\mathcal{I}}\omega_B)$ is the linear and angular velocity of the base frame w.r.t. the inertial frame, while $\dot{s}$ are the joint velocities.

The floating-base system equations of motions can be derived following the Euler-Poincaré formalism [15]:

$$M(q)\dot{v} + C(q, v)v + G(q) = \begin{bmatrix} 0_6 \\ \tau \end{bmatrix} + \sum_{k=1}^m J_k^\top F_k, \quad (1)$$

where $M, C \in \mathbb{R}^{n+6 \times n+6}$ are the mass and Coriolis matrix, $G \in \mathbb{R}^{n+6}$ is the gravity vector, $\tau \in \mathbb{R}^n$ are the internal actuation torques and F_k is the k^{th} external force acting on the system. In the case under study, we assume that the forces acting on the robot during flight are the aerodynamic forces and the thrust forces generated by the jet-engines. Here, we consider in the model only the thrust generated by the jet-engines, while the addition of aerodynamic forces in Eq. (1) will be treated later on in the paper. Hence, each $F_k \in \mathbb{R}^3$ represents the thrust force applied on the robot by the k^{th} jet, and we rewrite $F_k = {}^{\mathcal{I}}l_k T_k$, where ${}^{\mathcal{I}}l_k \in \mathbb{R}^3$ is the direction of thrust force, while $T_k \in \mathbb{R}$ is the thrust intensity. The

Jacobian $J_k(q)$ is the map between the system velocity v and the linear velocity ${}^x\dot{o}_k$ of the k^{th} thrust application point. We can then define the thrusts vector $T := (T_1, T_2, T_3, T_4)^\top$ and rewrite $\sum_{k=1}^m J_k^\top F_k = f(q, T)$.

C. Modeling of Aerodynamic Forces

We prefer to model the *global* aerodynamic force acting on the robot, because evaluating aerodynamic effects distributed on several parts of the body requires to design a more complex robot model for the CFD analysis, which becomes very time consuming. The global aerodynamic effects at the robot CoM are composed of the aerodynamic force $F_a \in \mathbb{R}^3$, and the moment $M_a := \mathcal{GP} \times F_a \in \mathbb{R}^3$, where $\mathcal{GP}$ is the distance between the CoM $\mathcal{G}$ and the Center of Pressure (CoP) P .

Consider any pair of angles (α, β) characterizing the relative orientation of v_a w.r.t. the body frame $\mathcal{D}$, chosen as in Fig. 2. $\alpha \in [0, \pi]$ is defined as the angle between v_a and the negative k axis of the body frame $\mathcal{D}$, while $\beta \in [0, 2\pi]$ is defined as the angle between the projection of v_a on plane (i, j) and the positive i axis of frame $\mathcal{D}$ (see also [14]).

The aerodynamic force can be decomposed w.r.t. the relative velocity frame $\mathcal{A}$ as:

$$F_a = F_D + F_S + F_N, \quad (2)$$

where:

- F_D , the drag force, aligns with Y_a axis;
- F_S , the defined sideforce, aligns with X_a axis;
- F_N , the defined normal force, aligns with Z_a axis.

The intensity of the steady aerodynamic force varies approximately with $|v_a|^2$, hence there exist three dimensionless functions C_D, C_S, C_N depending on the Reynolds number Re , the Mach number $\mathcal{M}$ and (α, β) , such that [14] [16]:

$$F_D := \mathcal{K}_a |v_a|^2 C_D(Re, \mathcal{M}, \alpha, \beta) j_a, \quad (3a)$$

$$F_S := \mathcal{K}_a |v_a|^2 C_S(Re, \mathcal{M}, \alpha, \beta) i_a, \quad (3b)$$

$$F_N := \mathcal{K}_a |v_a|^2 C_N(Re, \mathcal{M}, \alpha, \beta) k_a, \quad (3c)$$

where $\mathcal{K}_a := \frac{A_{ref}\rho}{2}$ with ρ the air density and A_{ref} the reference frontal area; i_a, j_a, k_a are the unit vectors of frame $\mathcal{A}$ along X_a, Y_a, Z_a axes. In particular, j_a has the opposite direction of v_a by definition. Identifying the numerical relation among the dimensionless coefficients C_D, C_S, C_N and their variables is fundamental for the design of analytical models of the aerodynamic forces to be in the system modeling and control design.

III. AERODYNAMIC COEFFICIENTS IDENTIFICATION

A. CFD Analysis

Computational Fluid Dynamics (CFD) analysis via the Finite Volume solver Ansys Fluent 2021 R1 is applied to estimate the aerodynamic forces acting on the studied robot platform (iRonCub). The robot shape may differ if the joint position changes. However, performing CFD simulations to cover all possible robot shapes in different flying conditions is time-consuming. An assumption is made to overcome this limit.

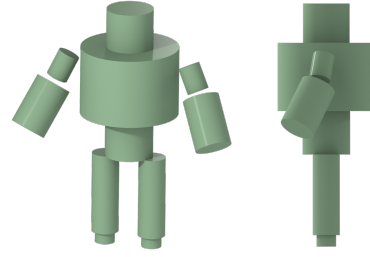


Fig. 3. Simplified iRonCub CAD model: initial joint configuration.

TABLE I
CFD SIMULATION SETTINGS

<i>Solver Settings</i>	Pressure-Velocity Coupling for RANS (Reynolds Averaged Navier-Stokes) Steady-State Simulation SST $k - \omega$ for Turbulence modelling No Compressibility Effects (i.e. $M \ll 1, \rho = \text{const}$)
<i>Constant Values</i>	Air density: $\rho = 1.225 \text{ kg/m}^3$ Reference pressure: $p_{ref} = 1 \text{ atm}$
<i>Boundary Conditions</i>	Inlet velocity: $v = 7.5 \text{ m/s}$ Outlet pressure: $p_{rel} = 0 \text{ Pa}$ Walls: No Slip Condition

Assumption 1: The robot shape variation due to joint movements is neglected considering that the robot joint position is kept close to the initial one by the flight controller during the entire flight envelope (see Sec. IV).

Considering **Assumption 1**, CFD simulations are all performed in a robot configuration corresponding to the initial joints position, which has been optimized for flight. Furthermore, a complex-shaped geometry as the iRonCub one can cause difficulties in the meshing phase of CFD simulations, because of errors caused by interferences between complex parts. Therefore, without losing the main geometrical characteristics of the robot, we designed an approximated CAD model of iRonCub which represents the major links as simple cylinders (see Fig. 3). The model allows to obtain high-quality meshes with a relatively low number of elements, which in turn translates in a reduction of the simulation time.

Settings of CFD simulations are schematically reported in Table I. Significant robot flight configurations, corresponding to precise values of α and β , were simulated using CFD tools. In total 21 simulations were performed (see Table II) which supplied the necessary data for proceeding with the modeling of aerodynamic characteristics on iRonCub.

B. Modeling Aerodynamic Characteristics

As introduced in Sec. II-C, the three dimensionless functions C_D, C_S, C_N are the aerodynamic characteristics of the body, i.e. the defined aerodynamic coefficients (AC): drag coefficient, sideforce coefficient and normal force coefficient. In this section, the analytical model of AC is identified based the CFD results from Sec. III-A. To proceed with identification, the following **Assumption 2** is made:

- 1) the aerodynamic forces act at the robot's CoM (co-

TABLE II
SIMULATIONS TABLE

$\alpha[^\circ] \backslash \beta[^\circ]$	180	225	270	315	360
15			✓		
30	✓				
45			✓		
60	✓	✓	✓	✓	✓
90	✓	✓	✓	✓	✓
120	✓	✓	✓	✓	✓
150	✓				
160			✓		
180			✓		

incides with CoP) and therefore the aerodynamic moments M_a are equal to zero;

- 2) furthermore, the sideforce effect is neglected due to its small magnitude from CFD results compared with the other two aerodynamic force components, and with other forces acting on the robot;
- 3) Mach number $\mathcal{M}$ and Reynolds number Re are assumed to be constant.

It is also possible to neglect the effects of rotational and unsteady motions of the vehicle on its surrounding airflow [16], and then, the AC depend only on the angle pair (α, β) and can be written as [14]:

$$\begin{aligned} C_D(\cdot) &= C_D(\alpha, \beta), \\ C_N(\cdot) &= C_N(\alpha, \beta). \end{aligned}$$

Considering that the aerodynamic coefficients are naturally 2π -periodic [20], sine and cosine functions are used to design the mathematical models of AC:

$$C_D(\alpha, \beta) = c_0 + c_1 \sin^2 \alpha \cos^3 \beta + c_2 \cos^3 \beta + c_3 \cos^2 \beta, \quad (4)$$

$$C_N(\alpha, \beta) = d_0 + d_1 \sin(2\alpha) \cos^2 \beta. \quad (5)$$

The coefficients characterizing the expressions of $C_D(\cdot)$ and $C_N(\cdot)$:

- $c_0 = 0.1327$, $c_1 = -0.0858$, $c_2 = 0.0679$, $c_3 = 0.0949$;
- $d_0 = 0.0042$, $d_1 = 0.0598$,

are obtained via Matlab Regression Learner App applying linear regression method and cross validation technique to avoid overfitting. Because in fluid dynamics drag force is a type of friction that acts opposite to the relative motion of any object moving w.r.t. the surrounding fluid, the estimated drag coefficient C_D should remain positive in the whole range of α and β according to our definition. This property is verified by the identified model in Eq. (4), as depicted in Fig. 4.

IV. CONTROL DESIGN

A. Flight Control Without Aerodynamics Awareness

The iRonCub flight controller [12], [13] is a *task-based* control strategy, composed of several control objectives (tasks), with different priorities. The so-called *primary task*, i.e. the task with the higher priority, is the stabilization of

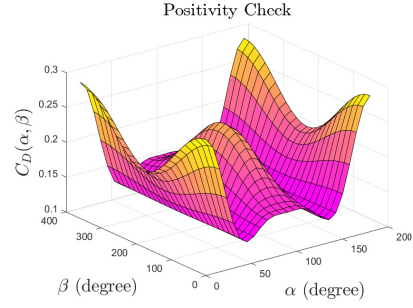


Fig. 4. C_D model described in (4). $\alpha \in [0, 180]$ deg and $\beta \in [0, 360]$ deg.

the robot's *centroidal momentum*² dynamics, which equals the summation of all external forces acting on the robot:

$$\mathcal{G}^{\mathcal{I}} \dot{h} = A(q)T - mge_3 = f_h(q, T), \quad (6)$$

where m is the robot's total mass, g is the norm of gravitational acceleration and $A(q) \in \mathbb{R}^{6 \times m}$ is a proper matrix mapping the thrust vector T in the momentum equation. For the momentum control design, we assume that the robot joints velocities $\dot{s}$ and the thrust rate of change $\dot{T}$ can be chosen at will, and then can be considered as our control inputs. Therefore, we define $u := (\dot{T}, \dot{s})$. To make the control input u appear in the momentum equation (6), we increase the relative degree of the output and write the momentum acceleration:

$$\mathcal{G}^{\mathcal{I}} \ddot{h} = \dot{A}(q)T + A(q)\dot{T} = f_h(q, T, v, \dot{T}). \quad (7)$$

As demonstrated in our previous works [12], [13], it is possible to find a smooth control input u^* that renders the closed-loop equilibrium point $(\tilde{h}, \tilde{h}, I) = (0, 0, 0)$ globally asymptotically stable, where $\tilde{h} = h - h^d$, h^d is the reference momentum trajectory and $I := \int_0^t \tilde{h}$.

We also define a *postural task* to resolve the redundancy in the control input u and maintain the robot configuration close to a desired shape:

$$\dot{s}^* = -K_P^s(s - s^d), \quad (8)$$

with s^d the desired robot posture and $K_P^s \in \mathbb{R}^{n \times n}$ a positive gains matrix. The two tasks are then achieved by means of Quadratic Programming (QP) optimization framework:

$$\begin{aligned} u^{**} &= \operatorname{argmin}_u (w_1 |u - u^*|^2 + w_2 |\dot{s} - \dot{s}^*|^2) \\ \text{s.t.} \quad & u_{\min} \leq u \leq u_{\max}, \end{aligned} \quad (9)$$

where *soft* task priorities are assigned to the two tasks via proper weights $w_1, w_2 > 0$. Framing the problem as an optimization procedure also allow to include input boundaries $u_{\min}, u_{\max}$ in the control design. It is also possible to include the integral of input boundaries (namely, the joint position and thrust limits) in the QP design. For more details on these aspects of the flight control design, which are not relevant for this paper, we refer to previous work [12], [13].

²the centroidal momentum is the robot total momentum expressed w.r.t. the frame $\mathcal{G}^{\mathcal{I}}$ defined in Sec. II.

B. Flight Control with Aerodynamics Awareness

The flight controller discussed in IV-A does not consider the effect of aerodynamic forces. This missing information may impair the performances of the control algorithm when aerodynamic forces are not negligible. To overcome this limitation, we implemented two modifications of the iRonCub flight controller: *feedback linearization* and *gain scheduling*. To implement our control strategy, we considered the following **Assumption 3**:

- we assume that $\dot{F}_a = 0$ in the calculation of the momentum acceleration.

Addition of Aerodynamics Forces: First, we include the estimated aerodynamic forces, obtained by writing Eq. (2) with the aerodynamic coefficients estimated in Sec. III-A, in the robot equations of motion Eq. (1) and momentum dynamics Eq. (6), which renders:

$$M(q)\dot{v} + C(q, v)v + G(q) = \begin{bmatrix} 0_6 \\ \tau \end{bmatrix} + f(q, T) + J_{CoM}^\top F_a, \quad (10)$$

where $J_{CoM} \in \mathbb{R}^{3 \times n+6}$ is the CoM Jacobian, while the momentum rate of change is now given by:

$$\mathcal{G}[\mathcal{I}]\dot{h} = A(q)T - mge_3 + \begin{bmatrix} F_a \\ 0_3 \end{bmatrix}. \quad (11)$$

Feedback Linearization Control: We assume that the relative velocity ${}^{\mathcal{I}}v_a$ can be measured, for example by employing dedicated sensors such as Pitot tubes applied on the robot, allowing the computation of the aerodynamic force F_a . A feedback linearization controller then substitutes Eq. (6) with Eq. (11) in the calculation of the baseline control input u^* , which depends on the momentum rate of change $\dot{h}$. In this way we can cancel out the effects of the aerodynamic forces on the desired closed-loop momentum acceleration.

Implementation of Gain Scheduling: Gain scheduling is employed as a strategy to react to wind gusts. In particular, we assume that a wind gust is detected when the computed aerodynamic force F_a overcomes a certain threshold and noticeably affects the robot momentum dynamics.

The desired closed-loop momentum acceleration has the following shape, which is obtained by applying the baseline controller u^* Eq. (9):

$$\ddot{h} = \ddot{h}^d - K_D\dot{\tilde{h}} - K_P\tilde{h} - K_I I. \quad (12)$$

When a wind gust is detected, Gain Scheduling smoothly modifies the value of the gain K_I , related to the integral of

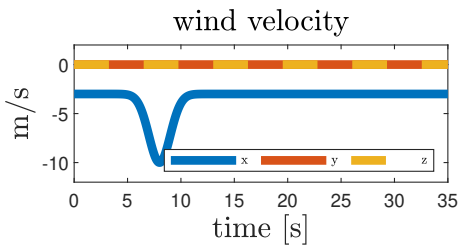


Fig. 5. Simulated air environment w.r.t. the inertial frame $\mathcal{I}$: ${}^{\mathcal{I}}v_w$, $\mathbf{X}$ blue line, $\mathbf{Y}$ red line, $\mathbf{Z}$ yellow line

the momentum error I , which depends directly on the CoM tracking error.

V. SIMULATION RESULTS

A. Simulation Environment

The robot dynamics is simulated by numerically integrating the equations of motion with aerodynamic forces Eq. (10) with the numerical integrator *ode15s* in a custom-made simulator implemented in MATLAB-Simulink. The controller is entirely designed in Simulink, and runs at a frequency of 100 Hz. A fast, inner torque control loop is designed to stabilize the joints velocities $\dot{s}$ towards the reference values generated with Eq. (9).

B. Flight Envelope Design

We designed a flight envelope composed of the following scenarios:

- 1) Scenario 1: the robot hovers in vertical position, in presence of a constant wind of 3 m/s in the frontal direction. After some time, a wind gust hits the robot still in the frontal direction. The wind gust is simulated by means of a Gaussian function as depicted in Fig. 5.
- 2) Scenario 2: the robot hovers and climbs to gain altitude, then moves forward at around 11 m/s, in presence of the same wind conditions presented in Scenario 1.

In both scenarios, the magnitude of the wind gust is assumed to be 10 m/s, and its duration is 5 s. These values were selected after analyzing wind data collected by means of a weather station, located close by the area where experiments with the real iRonCub occur. Fig. 7 shows the robot flying in the two scenarios described by the flight envelope.

C. Robustness Analysis with No Aerodynamics Awareness

We first evaluate the performances of the baseline controller presented in [12] [13] for the designed flight envelope. The controller has been fine tuned to achieve the best performances possible without aerodynamics effects.

Simulation results are presented in Fig. 8. We focus our attention on the CoM and linear momentum error norm, which are significantly affected by the disturbance introduced by the wind. As depicted by Fig. 8(a), the robot is still able to hover despite the presence of the wind. However, the CoM error norm considerably increases, with a peak of 30 cm error and a constant error of about 3 cm. For what concerns instead the high-speed flight scenario, for which robot control is more

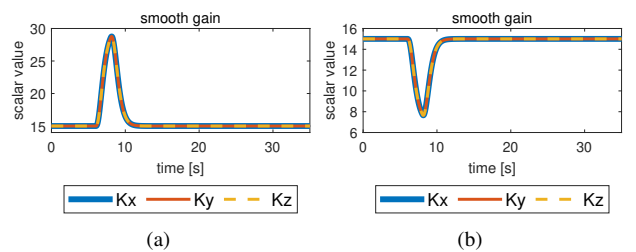


Fig. 6. Gain-scheduling: a) for hovering; b) for high-speed flight. $K_x, K_y, K_z \in \mathbb{R}^+$ are the elements along the diagonal of matrix K_I .

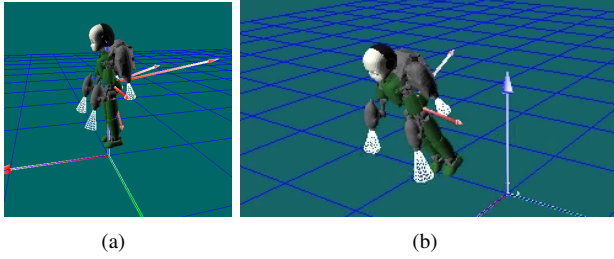


Fig. 7. a) Scenario 1; b) Scenario 2, with the aerodynamic force vector and body frame shown as arrows on the body.

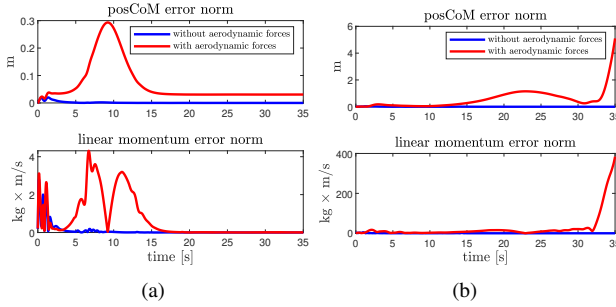


Fig. 8. a) Hovering scenario: flight controller that does not account for aerodynamics, with (red line) and without (blue line) aerodynamic effects acting on the system; b) High-speed scenario: flight controller that does not account for aerodynamics, with (red line) and without (blue line) aerodynamic effects acting on the system.

critical because the system dynamics are more excited, the controller is not able to retain stability in presence of wind, and the robot crashes on ground, as shown by Fig. 8(b).

D. Control Results with Aerodynamics Awareness

The two scenarios in the flight envelope are then tested with the flight controller modifications proposed in Sec. IV. Results are presented in Fig. 9, that show the CoM and linear momentum error norm during the flight simulations. In the hovering scenario depicted in Fig. 9(a), the feedback linearization controller (red line) manages to considerably reduce the CoM error despite the external disturbances. However, the closed-loop system dynamics presents oscillations, visible in particular in the linear momentum error plot in Fig. 9(a). To fairly compare the results of feedback linearization controller with the baseline controller previously developed, we did not perform any change in the controller's gains: we believe that this oscillation could be mitigated by re-tuning all the gains for this new controller. On the other side, gain scheduling helped to reduce the maximum CoM position error by $\sim 38\%$ w.r.t. the baseline controller during the wind gust.

In the high-speed scenario of Fig. 9(b), both controllers are able to retain system stability despite the presence of wind. Feedback linearization control also allows to maintain small CoM and linear momentum error norm during the whole simulation. In gain scheduling case, we decided to reduce the CoM gains while the wind gust occurs. The gains relaxation allows to preserve robot control at the cost of a large errors on the CoM position. Fig. 6 shows how the elements of the gains matrix K_I vary for both reference scenarios.

TABLE III
GAIN-SCHEDULING V.S. FEEDBACK LINEARIZATION

	Max-pos-err: hovering	Max-wind-gust: high-speed flight
Current controller	0.29 m	9 m/s
Gain-scheduling	0.18 m (-38%)	10 m/s ($+11\%$)
Feedback linearization	0.035 m (-88%)	13 m/s ($+44\%$)

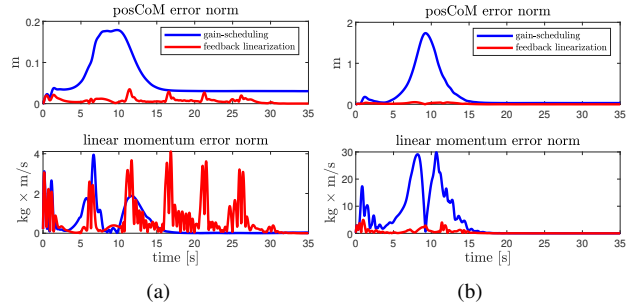


Fig. 9. a) Hovering Scenario: gain-scheduling (blue) V.S. feedback linearization (red); b) High-speed scenario: gain-scheduling (blue) V.S. feedback linearization (red).

The performances of the baseline and aerodynamics-aware controllers are briefly compared and summarized in Table III, and a video describing the performed simulations is attached to the paper.

VI. CONCLUSIONS

In this paper, analytical models of aerodynamic forces acting on the jet-powered humanoid robot iRonCub are identified using data coming from CFD analysis. The identified models are then used to design appropriate controllers for the robot flying in the given flight envelope. Two techniques, *gain-scheduling* and *feedback linearization*, are applied to improve the flight controller in presence of aerodynamic effects. The controller was tested successfully through simulations in a critical outdoor environment.

Nevertheless, the proposed framework still requires several assumptions, both for performing the CFD analysis and in the controller design. Future work may consider to distribute the aerodynamic forces all over the major robot links for more realistic simulations, and improve the geometry of the iRonCub model for CFD analysis. Finally, further effort in control design is desired to improve the robustness of the controller in different flight envelopes.

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